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Crowdsourced Civic Issue Reporting And Resolution System

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Abstract**—** A Cityes and rural areas often encounter civic issues (for example potholes, garbage buildup, leaking plumbing, malfunctioning streetlights, and drainage problems), which affect the quality of life of citizens in both cases. Traditional reporting mechanisms for civic issues have relied on manual complaints, protracted processes with little transparency and slow resolution times. Thus, for a number of reasons, many civic complaints go unresolved or are otherwise handled poorly. To resolve these problems, a Crowdsourced Civic Issue Reporting and Resolution System is being implemented. It is a technology-based solution that will help to increase citizen participation in local government and to resolve civic issues.The Crowdsourced Civic Issue Reporting and Resolution System allows citizens to report civic issues in real-time on digital platforms (for example, mobile and web applications). Citizens can submit complaints using digital images, by entering descriptive information about the issue and, by using GPS coordinates to report the exact location of the issue. Once a citizen submits a complaint, the civic issue will automatically be categorized and forwarded to the appropriate municipal department. This process minimizes the amount of manual effort required to resolve civic issues by ensuring that all complaints are delivered to the relevant municipal department in a timely manner.

I. INTRODUCTION

As cities and towns flood with people and grow, urbanisation leads to increases of demand for civic authorities' services in these places, and citizens face many of their cities' basic civic issues every day (broken roads, overflowing garbage bins, broken water supply, broken street lights, open manholes, clogged drains). These problems create inconveniences for citizens but also pose real threats to their health, safety, and overall quality of life. Without solutions to civic issues and processes, sustainable development of cities cannot happen. The traditional methods of reporting civic issues such as submitting a complaint to a municipal office, writing a letter to complain, or calling in to report an issue, are slow, cumbersome, and inefficient. In addition, citizens have little or no confidence that their complaint has been resolved satisfactorily or that local government is dealing with it in a timely manner. Because citizens cannot easily check the status of their complaint, they become frustrated with the process, which leads to further erosion of their trust in local government. Because of the amount of time it takes to resolve complaints and due to poor communication (between the departments), many complaints go unresolved or lost, and the result is no resolution due to a lack of coordination and communication among municipal departments. Technological advances and the growth of mobile technology are creating more effective ways for citizens to document and report civic issues throughout their cities and towns. An online civic issue reporting and resolution system (Crowdsourced) allows residents to quickly easily communicate with the city's government officials about civic problems in the civic region or zone where they live. Residents can submit civic issues they encounter through a web or mobile application from anywhere without needing to meet with an official face-to-face. However, when reporting an issue via the app, residents are encouraged to upload images, descriptions and/or GPS locations that will positively and negatively affect accuracy of the report and resolving the issue will be streamlined due because of the detailed accuracy of the report.

The system promotes strong community participation and involvement in addressing civic issues because it provides opportunities for citizens to validate issues that have been reported by providing their support for the issue. Citizens can rally support from other citizens who can vote on, comment on or provide other forms of support in favor or against the reported issue. The community works collectively to identify, validate and agree on issues that should receive the highest priority for resolution as work as a public group.

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3

II. LITRATURE SURVEY

.The number of people in cities and urbanized areas is rapidly increasing leading to many urban centers struggling with managing residents and service delivery. Governments at the local level are facing incredible challenges due to an increase in residents and the significant stress they are placing on the cities they reside in... In the last few years, several research projects and practical implementations have been completed that demonstrate the very real need for technology to be incorporated into civic reporting processes to make reporting and resolving civic issues more efficient. Khalil et al., (2014) suggest that traditional means of receiving complaints from citizens (in the form of paper-based forms, via telephone, and in-person) are near impossible to use effectively, as they do not allow for time sensitive reporting, are often delayed, and do not provide citizens with accountability and/or transparency regarding their claims. Consequently, the inefficiencies and ineffectiveness result in a general dissatisfaction of citizens and ultimately cause an erosion of trust in their local government. Based on this evidence, researchers have focused on creating technology delivery systems to enable/speed up the real-time submission of civic issues in order to facilitate communication between citizens and cities. Crowdsourcing has become one of the most researched approaches within this area and is defined, by Howe (2006).Research on Civic Reporting Systems has progressed throughout time. The purpose of many studies has been to create and implement a civic reporting system. For example, Xiong et al. (2015) examined a mobile-based report submission platform that allowed citizens to submit geo-tagged reports of local issues, such as damaged physical infrastructure. Their results indicated that geo-tagged reports led to quicker responses from city authorities than non-geo-tagged reports, since the city authorities were able to locate the issue more easily.Longitudinal studies have also examined how public feedback mechanisms such as upvoting, commenting and validation by the community affect the civic reporting systems that are in place. Chen et al. (2018) found that public engagement mechanisms have facilitated decision making for authorities regarding prioritization of issues based on the collective judgement of the public. In other words, if an issue receives more public support, the issue will generally be critical and receive resources from the authorities to resolve it compared to issues that received very little public support. Gupta and Kumar (2019) also indicated that the use of social validation decreases the number of instances of duplicate reports, and improves the quality of the data collected through clustering similar types of complaints.

III .PROPOSED SYSTEM

The introduction of the new system, everyday citizens can submit their concerns about municipal/city infrastructure problems via a mobile app or a web portal. This should facilitate better communication between municipal authorities and their constituents through a more transparent and structured method of tracking complaints in real time.Upon registration and login into the system, users will be able to submit a complaint about various municipal infrastructure issues, including potholes, garbage overflow, water leaks, broken street lights, clogged drains, etc. Users will have the ability to upload an image, provide a detailed description of the complaint, and to grant the system access to their GPS location, so that the exact location of the complaint is known. By using these three data points (image, description, and GPS location), the system seeks to provide an accurate record of the location of the complaint, thus removing the potential for citizens and municipal officials to be unclear as to where the problem exists.

Each complaint received by the system will be stored in a central database with an assigned complaint number for tracking. The system contains an automated processing module that will classify complaints into pre-defined categories. Based on the classified categories and the designated geographic location, the system will automatically route the complaint to the appropriate municipal department. The automated complaint routing is designed to eliminate any delays caused by the manual routing of complaints.The proposed administrative dashboard allows municipal officials to access any/all incoming complaints, filtered by type/category, location, and priority level. The municipal officials will then assign the complaints to municipal field workers for completion, while simultaneously monitoring their real-time status information in order to ensure timely action.Municipal field workers will automatically receive notification(s) of assigned tasks; and can update the complaint's status (e.g., disposition of "Received," "In Progress," and/or "Resolved") upon completion and/or inspection of the particular complaint(s). Correspondingly, citizens will receive notification (via SMS, e-mail, or in-app notification) of any/all updates made to their complaint(s) status.

The proposed system will also provide an avenue for citizens to rate/responses regarding the quality of service they received in resolving their complaint; as well as providing response suggestions. This will help to create accountability for municipal departments, thereby enhancing overall level of service delivered to municipal citizens.Additionally, the proposed administrative dashboard includes data and analytic tools for analyzing trends and identifying locations

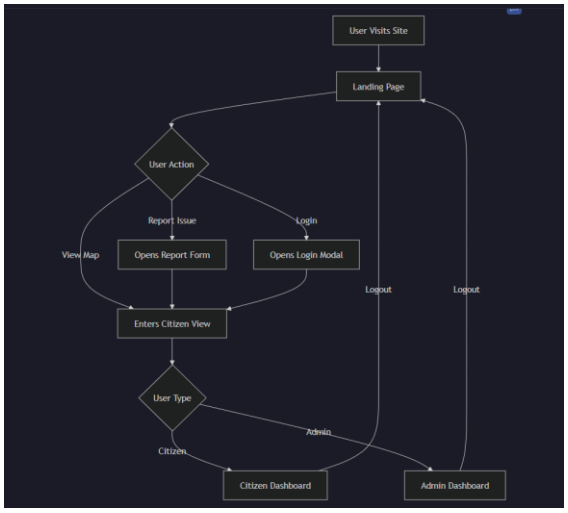
where complaints are frequently reported; this ultimately assists municipal officials in taking proactive approach as well as improving municipal infrastructure management. Furthermore, the identified trends and patterns will assist municipal staff in developing heat maps and/or graphical representation(s) of issues prevalent across the various geographical regions of the municipality.

- **FRONT END** - The Crowdsourced Civic Issue Reporting and Resolution System has a front end that allows users to interact with the system. The user interface was developed using HTML5 for structure and CSS3 for layout. These two languages help give the application a visually appealing user interface and also enable the application to function correctly across different types of devices.JavaScript provides the pages with added functionality and will dynamically change their behavior based on user input and will also improve the performance of modern-day front-end frameworks such as React.js and Angular. These frameworks will allow you to develop an SPA to improve your application's performance, as well as provide a better user experience.Applications that use responsive design use frameworks such as Bootstrap or Tailwind CSS to cater for different types of devices . If the final application is going to be developed as a mobile application using cross-platform technologies such as Flutter or React Native, the final product will contain one code base for both Android and iOS applications.To integrate maps into the application front end, the use of the Google Maps API or the Open Map API was used. Users will be able to locate and help other users identify where they need to go by displaying common points of interest (POI).

- **BACK END** - Backend systems are designed to support the development of essential functional capabilities with respect to data processing (including user authentication), databases (such as Azure, Google Cloud, and Amazon Web Services (AWS) used for Back and data to host, process, scale, and store behind the scenes for all applications).Backend systems are constructed using one of several common programming languages, e.g., Java (Node.js), Python, and/or Java.

Backend systems often leverage frameworks (Django, Flask - Python, Spring Boot - Java, and Express.js - Node) to create a RESTful API for the backend server application. As a rule of thumb, an individual developer who uses one of those frameworks will build a backend server that performs the following tasks: processes authentication, processes and categorizes submitted complaints, provides the user with status updates on their complaints, and provides administrative access to the processed authentication, processed and categorized submitted complaints, and the information on the complaint status.Most backend databases are relational databases (e.g., MySQL, PostgreSQL) because they require a structured format for persistent storage (e.g.,) of user information, submitted complaint records, and the history of how submitted complaints were resolved. However, a no-SQL database such as MongoDB may also be used when the need for a more scalable and flexible model of persistent storage exists.

V ARCITECTURE



civicsense is founded on the three-tier organization:
presentation layer react spa (typescript) on tailwind css and leaflet.js, providing citizen interface of reporting and management dashboard of administration.
application layer fastapi (python) server which includes authentication, business logic, ai coordination with gemini integration, image processing.
data layer: supabase (postgresql) database including user profiles, reports database and image store including row-level security policies.
the interaction between the presentation and application layer, the application and data layer is carried out by using the rest api (https) and sql queries.

VI. RESULTS AND DISCUSSION

Crowdsourced Civic Issue Reporting and Resolution Systems lead to a highly efficient, transparent, and citizen-friendly way to manage civic complaints. The system allows citizens to report a variety of civic complaints, such as potholes, overflowing rubbish bins,

leaking water, and broken street lights via their mobile device or via web application.

Once the system is deployed, citizens can quickly file complaints with real-time registration of their geographical location via GPS integration. Each complaint issued receives a unique identification number so users may track the progress of their complaint from when they submit their complaint until it has been completely resolved. This is a marked improvement over traditional manual processing of complaints where no tracking could take place.

The automatic categorization and routing of complaints guarantees that complaints are immediately directed to and received by the appropriate municipal department. As a result, response times are decreased, and the resolution of issues becomes more formalized and organized. Moreover, management can effectively delegate tasks to municipal workers in the field and track their progress through an administrative dashboard.

The use of crowdsourcing allows multiple people (users) to view, support, or up-vote any civic complaint listed within the system. This also allows for the prioritization of critical complaints based on community interest and impact. As a result, the highest-priority complaints receive more immediate responses than those with lower-priority complaints and result in better allocation of municipal resources and improved citizen satisfaction.

VI.CONCLUSION AND FUTURE WORK

A new way to connect citizens with their governments through allowing a crowd-sourced way of working on solutions to the inherent flaws within traditional systems of accepting complaints. Providing citizens with a civic-oriented, digitally accessible way to submit their issues via a fully digital process creates an opportunity for active involvement by people in the improvement of the communities where they live.The use of GPS location information to help accurately track complaints, photographic uploading for immediate identification of problem detail, auto-categorizations for easier tracking, and real-time status updates should improve the accuracy of complaint handling, increase efficiency, and create greater accountability for complaint handlers. This should all improve the applicant's experience when using the complaint mechanism.Local government administrative staff should be able to utilize centralized dashboards to oversee and manage their workload through completion of numerous complaints that were submitted. The use of this crowdsourcing capability provides citizens with another way to provide feedback on the types of issues that they consider most critical and creates a mechanism for developing a prioritization system that assists government agencies in determining which complaints need to be investigated first. By utilizing data analysis and visualization tools, local governments will be able to identify trends in the types.

The existing program does a great job with the process of managing citizens' concerns surrounding civic-related issues; however, there are many steps that could be taken to improve the program in order to enhance efficacy and overall functionality moving forward.One planned improvement is the addition of AI (artificial intelligence) and ML (machine learning) algorithms that will automatically categorize citizen complaints more accurately and generate predictive analytics around historical data to identify areas at the highest risk of future civic problems. This will greatly assist municipalities in taking a proactive approach to maintenance instead of having to react to problems once they have occurred.Another planned upgrade is the addition of live chat between citizens and municipalities providing instant clarification on issues and providing faster communication. Additionally, adding a chat bot feature will allow municipalities to provide automated responses to the majority of citizen inquiries.The system would also be improved by adding the ability to submit complaints through voice commands, using multilingual voice-based reporting tools, which will help to reach the residents with low levels of literacy and/or technical experience who may not have access to the system.Additionally, the new versions of the program may include the capability to incorporate IoT (Internet of Things) sensor integration (smart waste bins and water leak detection sensors) that would allow the program to automatically detect and report problems to municipalities without the need for manual intervention.

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